Project Management Plan

for

<Akita>

Version 1.0 draft 1

Prepared by <Team Luminous>

<Cpt_S 484, Washington State University >

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Revision History

Name	Date	Reason for Changes	Version
Runlin Zhao	9/12/2026	Overview and Section 1	1.0 draft 1
Nathaniel Gaines	9/16/2026	Section 2-3.2	1.1 draft 2
Nathaniel Gaines	9/20/2026	Revising sections 2-3.2	1.2 draft 3
Runlin Zhao	10/1/2026	Remove the template text	1.3 draft 3

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1. Overview

This section of the Software Project Management Plan (SPMP) provides an overview of the project. It outlines Project Akita's primary motivation, objectives, success criteria, major deliverables, and constraints. The following subsections detail the foundational business needs, the specific scope of the project, and the critical parameters that will guide the design, requirements engineering, and prototype implementation of this indoor navigation system.

1.1. Project Purpose, Objectives, and Success Criteria

The primary purpose of Project Akita is to develop an indoor navigation software system designed to assist blind individuals in safely navigating complex indoor domains, including multi-floor environments with classrooms, offices, and elevators. The system must calculate optimal routes and deliver precise, voice-guided instructions. To accommodate users who may lack spatial awareness of standard metrics (such as children or those with no concept of meters), the application will personalize navigation by utilizing the user's inputted height and weight to calculate an estimated stride length, translating distances into exact step counts (e.g., "walk for 30 steps").
Furthermore, the system must utilize advanced smartphone sensors to detect obstacles for collision avoidance and initiate emergency communications if a fall is detected or location tracking is lost. Safe, fast, and comfortable navigation are the highest non-functional priorities, requiring highly customizable settings such as adjustable volume and instruction intervals to ensure optimal usability. Project success will be determined by the delivery of complete, unambiguous requirement engineering artifacts and a functioning prototype that demonstrates these core "TO-BE" scenarios. Other specific issues to address might include:

- Shared resources and their availability: Utilization of generative AI tools for rapid prototype development and reliance on standard smartphone hardware sensors for testing.
- Shared designs, code, and hardware components: Integration with existing smartphone accessibility APIs (e.g., VoiceOver, TalkBack) and built-in sensors (accelerometer, gyroscope, microphone).
- Feature dependencies: The personalized step calculation feature is heavily dependent on the accurate initial input of the user's biometric data (height/weight) and the reliable calibration of the device's internal pedometer.
- Schedule dependencies: Delivery of the functioning prototype and requirement artifacts is strictly constrained by the project timeline, requiring phased milestones for requirement gathering, design, and GenAI-assisted implementation.

1.2. Project Deliverables

The primary deliverables for Project Akita include comprehensive management planning documentation, requirements engineering artifacts, structural design specifications, and a functioning software prototype. To ensure an adequate basis for continuous design and testing,

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any currently missing information or scheduling details are marked as TBD (To Be Determined) until finalized with stakeholders.

Deliverable	Delivery Date	Delivery Method	Comments
Software Project Management Plan (SPMP)	TBD	Digital Document	Specifies the organizational structure, process model, schedule, and resource allocation for the project.
Software Requirements Specification (SRS)	TBD	Digital Document	Details all functional and non-functional requirements, ensuring verifiability, correct implementation priority, and full cross-reference traceability.
Software Design Description (SDD)	TBD	Digital Document	Describes major architectural components, visual models, internal interfaces, and expected error condition behaviors.
Functioning Prototype	TBD	Executable / Code Repository	A functional application demonstrating the proposed "TO-BE" scenarios, including the personalized step-calculation algorithm and voice-guided indoor routing.

1.3. Assumptions, Dependencies, and Constraints

Assumptions

- **AS-1:** The target demographic (blind users) or their caretakers have access to advanced smartphones equipped with functional hardware sensors (accelerometer, gyroscope) and voice recognition capabilities necessary to support the application.
- **AS-2:** Users or their caretakers can accurately input personal biometric data (height, weight, and age) to facilitate the customized step calculation algorithm.
- **AS-3:** The requirements documented throughout the project life cycle will represent actual system needs, rather than premature solutions or implementation constraints.

Dependencies

- **DE-1:** The successful implementation of precise obstacle avoidance algorithms is heavily dependent on the high sensitivity and accuracy of the smartphone's built-in hardware sensors (e.g., gyroscope, accelerometer, camera, or LiDAR) to reliably map physical surroundings and calculate collision risks.
- **DE-2:** The automated emergency response system relies on seamless integration with local medical dispatch networks (e.g., 911) and predefined

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secondary stakeholders (e.g., family members or caretakers). Upon detecting a fall or severe anomaly, the application depends on user-configured safety protocols; if the user fails to respond to a severity-assessment prompt within a customized timeframe (e.g., 15 or 30 seconds), the system must successfully connect to cellular networks to automatically dial 911 and transmit the user's exact coordinates.

- **DE-3:** The successful delivery of the functioning prototype within the constrained project timeline is dependent on the team's ability to efficiently leverage modern software development frameworks to translate requirements engineering artifacts into executable scenarios.

Constraints

- **CO-1**: The system's operational domain is strictly confined to indoor environments, such as multi-floor buildings containing classrooms, offices, and elevators.
- **CO-2**: Time-critical functions, specifically obstacle detection and collision avoidance warnings, are heavily constrained by strict timing criteria to ensure safe navigation.
- **CO-3**: Every defined requirement must be technically feasible, implementable within known constraints, and verifiable by testing, demonstration, review, or analysis.

Project Dimensions Analysis

- **Constraints (Fixed):**
  - **Schedule:** The delivery date for the requirement engineering artifacts and the functioning prototype is strictly bound by the academic semester timeline.
  - **Staff:** Development resources are limited strictly to the assigned undergraduate team members (Team Luminous).
- **Project Success Drivers:**
  - **Quality:** Because the primary users are blind, quality attributes—specifically safety, usability, and comfortable navigation—are the paramount drivers of project success and must be measurable.
- **Degrees of Freedom (Flexible):**
  - **Features:** While core indoor navigation and obstacle detection are mandatory, advanced predictive features (e.g., suggesting destinations based on a user's routine habit) can be scaled back or identified as TBD if they conflict with schedule constraints.

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- **Cost:** The project operates with a highly flexible, near-zero budget, utilizing free or existing student resources and GenAI tools.

1.4. References

- ***IEEE Std 1058-1998:*** *IEEE Standard for Software Project Management Plans, IEEE Computer Society.*
- ***Project THEIA Preliminary Definitions:*** *Cpt_S 484 Project THEIA Preliminary Definition - v1.0, Washington State University.*
- ***Requirements Review Checklist:*** *Review Checklist for Software Requirements Document.*
- ***Client Correspondence:*** *Email Communication, "THEIA Project Clarification on Future Implementation and App/Model Requirements," by Bolong Zeng, Ph.D., Associate Professor, Washington State University.*

1.5. Definitions and Acronyms

Akita: The designated software project name developed by Team Luminous. The name is inspired by the beloved dog of Helen Keller, the author of *Three Days to See*, symbolizing loyalty, guidance, and trusted companionship for the visually impaired.
THEIA: The original domain project name specifying the indoor navigation requirements for blind users.
GenAI: Generative Artificial Intelligence, utilized in this project to assist in rapidly developing the functioning prototype.
Secondary Stakeholders: Individuals who assist the primary blind user by configuring the app settings or providing emergency aid, such as caretakers, family members, accessibility staff, or police.
SPMP: Software Project Management Plan. The document defining the process model, organizational structure, schedules, and resource allocation.
SRS: Software Requirements Specification. A document clearly detailing the essential functions, performances, external interfaces, and design constraints of the software.
SDD: Software Design Description. A document describing the major architectural components of the software design, including databases and internal interfaces.
TBD: To Be Determined. Used to explicitly identify needed information that is currently missing from the requirements.
TO-BE Scenarios: Future-state user workflows and process flows that the functioning prototype is expected to demonstrate.

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2. Project Organization

2.1. Process Model

Team Luminous will follow an incremental process model, planning the complete system while building only a core subset of it during Phase I. With three team members, no budget, and a single semester timeline, this is the most realistic approach. It also matches Dr. Zeng's guidance from his 9/4/2026 email, in which he clarified that Phase I is meant to demonstrate an understanding of the requirements rather than deliver a fully working app.

Core scope for Phase I: location-based navigation between two simplified test environments, using large buttons and voice interaction, with the user's starting location assumed rather than automatically detected. This scope is demonstrated through the prototype mock-up and the TO-BE scenarios.

Extended scope: the team will document the items below but is not committing to build them for Phase I. They will be evaluated only if time allows.

- Object and obstacle recognition using an existing pre-trained model such as YOLO, applied to a pre-recorded demo video.
- GPS or Wi-Fi based automatic positioning.
- Distinguishing temporary obstacles from permanent ones.

The team's Phase I phases are:

Phase	Key Tasks	Target dates
1. Requirements elicitation and issue analysis	Review the Preliminary Definition, identify issues, and document how each is resolved and why (WRS Issues)	9/21 to 10/6
2. Plan and specification refinement	Revise the plan using Dr. Zeng's feedback; develop the Improved Understanding (Problems, Goals, Objectives, Requirement Specifications, traceability)	10/11 to 10/13
3. Scenarios and prototype	AS-IS/TO-BE scenarios, presentation slides and recording, prototype mock-up, preliminary user manual	10/5 to 10/16
4. Final review and submission	Teamwork report, full team review, repository check, submission	10/16 to 10/18

Phase II follows the same approach, with a draft due 11/8/26 (est.) and the final due 12/6/26 (est.)

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2.2. Organizational Structure

Team Luminous consists of three members: Runlin Zhao, Keagen Chapman, and Nathaniel Gaines. The team uses a flat structure. All three members contribute to both requirements work and prototype work, since the team's size and budget do not allow for separate specialized roles. Each deliverable has one point person who is accountable for it (see Section 2.3). Although the team intends to make major decisions together we have elected Runlin as the team leader. Our day-to-day coordination happens on the team's Discord channels, and project documents are kept in the team's GitHub repository and Google Docs.

2.3. Roles and Responsibilities

Three coordination roles keep the work organized. They do not replace ownership of deliverables.

Role	Person	Responsibilities
Communication Liaison	Runlin Zhao	Main point of contact with Dr. Zeng for official communication and scope clarification.
Meeting Coordinator	Nathaniel Gaines	Schedules meetings, tracks agenda items and task status in between.
Technical Research Lead	Keagen Chapman	Leads research into technical tools and feasibility (YOLO, OpenCV, TensorFlow) for the demo and brings findings back for the team to discuss.

Each Phase I deliverable from Project Specification I has one point person, who drafts it, keeps it on schedule, and coordinates its review. The supporting members contribute content and review the deliverable before its review date.

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Deliverable (spec weight)	Point person	Supporting
WRS: Issues (25%)	Runlin Zhao	Nathaniel Gaines, Keagen Chapman
WRS: Improved Understanding (25%)	Nathaniel Gaines	Runlin Zhao, Keagen Chapman
AS-IS/TO-BE scenarios, slides, and recorded presentation (20%)	Keagen Chapman	Nathaniel Gaines, Runlin Zhao
Prototype mock-up and preliminary user manual (10%)	Keagen Chapman	Nathaniel Gaines
Project Plan, maintained through Phase I (10%)	Nathaniel Gaines	Runlin Zhao
AI usage and teamwork report (10%)	Runlin Zhao	Nathaniel Gaines, Keagen Chapman

This split puts roughly one third of the graded work on each member: Runlin 35%, Nathaniel 35%, Keagen 30%. It matches the equal-contribution expectation in CO-4.

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3. Managerial Process Plans

3.1. Management objectives and priorities

Team Luminous's main management priority is to produce complete, well-reasoned planning and requirements documentation for the full system, while limiting Phase I prototype work to a mock-up that demonstrates the core scope in Section 2.1. This follows Dr. Zeng's guidance and the project specification, in which the WRS document accounts for 50% of the Phase I grade and the prototype and user manual account for 10%. Extended features are treated as optional and labeled as such, not as committed deliverables. As established in Section 1.3, quality is the team's main success driver, schedule and staffing are fixed, and feature scope and cost are flexible.

Additional priorities:

- **MO-1:** Even distribution of work among team members, with clear task ownership and accountability (Section 2.3).
- **MO-2:** Direct communication with Dr. Zeng, through the Communication Liaison, whenever requirements or scope are unclear.
- **MO-3:** Weekly synchronization on the team's Discord channels between in-person meetings.
- **MO-4:** Transparent AI usage. Team members log any use of generative AI tools, with screenshots, and review the output before it enters a deliverable, so the AI usage and teamwork report is complete.
- **MO-5:** All Phase I work products are kept in the team's GitHub repository.

3.2. Assumptions, dependencies, and constraints

Assumptions

- **AS-4:** The team assumes Dr. Zeng's guidance that Phase I does not require a working prototype remains accurate, though he has indicated expectations may become clearer after Phase I is complete.
- **AS-5:** The team assumes continued free access to tools such as YOLO, OpenCV, and TensorFlow if any extended scope demo work is attempted.

Dependencies

- **DE-4:** The team depends on Dr. Zeng's feedback on this preliminary plan, especially on the timeline and responsibility breakdown, to revise the schedule before the final Phase I submission on 10/18/26.
- **DE-5:** Meeting scheduling depends on the availability of campus meeting rooms, which the team found difficult to book on short notice on 9/14. Discord meetings serve as the fallback.

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- **DE-6:** Final scope decisions on image recognition, temporary versus permanent obstacles, and GPS-based tracking depend on Dr. Zeng's response to the team's scope clarification questions.

Constraints

- **CO-4:** The team is constrained by the course's 60/40 peer review grading split, which requires roughly even individual contribution, about 33% per member on this three-person team. Clear task ownership in Section 2.3 supports this.
- **CO-5:** Phase I work is limited to the deliverables in Project Specification I. Items a commercial project might include, such as design documents or test plans, are out of scope.

3.3. Risk management

The primary method by which risks will be assessed is by having the team member responsible for a task report on risks related to the task. As part of assigning tasks, they will be assigned the associated risks to monitor. When a risk is identified, team members will report to the project manager, who will decide what action to take in response. Risk management is planned to take 5% of project hours.

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4. Technical Process Plans

4.1. Methods, tools, and techniques

The project is intended to operate on Android and iOS. The code will be hosted on github. Team communication will be conducted on discord. The primary language will be python for its extensive libraries and ease of implementation. The IDE that will be used is Visual Studio Code. The coding process will be conducted as TDD (Test Driven Development).

4.2. Software documentation

Document	Template or Standard	Target Date	Distribution
Project Phase 1 Draft	WRS	9/20/26	Google Docs
Project Phase 1	WRS	10/18/26	Google Docs
Project Phase 2 Draft	WRS	1/17/27	Google Docs
Project Phase 2	WRS	2/21/27	Google Docs

Appendix Table 1

Risk	Target Date	Type	Likelihood	Description
Incompatible tools and libraries	11/30/26	Technical	Medium-High Impact	Tools may be incompatible or require different versions of the same software
Failure to meet deadlines	All deadlines	Managerial	Low-High Impact	Deliverables are not completed on schedule
Scope Creep	12/11/26	Managerial	Medium-High Impact	Repeated increase in scope of project
Information Loss	Continuous	Technical	Low-High Impact	Documentation, work, and sources are lost, corrupted, damaged, or otherwise unavailable to the team

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